Few-Shot Semantic Segmentation on Remote Sensing Images With Learnable Prototype

Jing Wang, Yuang Liu, Qiang Zhou, Zhibin Wang, and Fan Wang

***Abstract*— Deep learning-based semantic segmentation has been the dominant solution to quickly capture regions of interest (ROIs) in remote sensing (RS) images. However, the annotation and training cost of a fully supervised segmentation model is often too high due to the requirement for elaborate masks. Additionally, trained models are limited to recognize only those classes defined in the training set. This has led to increased interest in how to cheaply adapt learned knowledge to new unseen objects. In this article, we propose a meta-learning-based few-shot method called learnable prototype few-shot segmentation (LPFS) to quickly adapt models to previously unseen geographic categories with only a few support examples of RS images. Specifically, we first build a learnable prototype module (LPM) based on variational autoencoder (VAE) to eliminate interclass ambiguity and extract high-level semantic prototypes from the support set effectively. We then design a global-attention correlation map to achieve low-level structural feature alignment between the support and query images. Additionally, we introduce a base learner to alleviate the bias caused by the meta-learning network on base classes. The extensive experiments on the public few-shot segmentation benchmark iSAID-5^i demonstrate that our method sets a new strong baseline for few-shot semantic segmentation on RS images.**

***Index Terms*— Few-shot learning, learnable prototype, remote sensing (RS), semantic segmentation, variational autoencoder (VAE).**

I. Introduction

With the development of modern remote sensing (RS) satellites, RS images with increasing resolution have been widely used in natural disaster monitoring, urban planning, change detection, and so on. Extracting regions of interest (ROIs) from those high-resolution RS images rapidly becomes an urgent demand. Recently, deep neural networks (DNNs) have achieved a great success in RS image segmentation [1], such as crop segmentation [2], [3], [4], urban scene parsing [5], [6], and so on [7], [8], [9].

Ordinarily, the astounding performance of DNNs is driven by the numerous data with delicate labels, however, the cost of data collection and label annotation is exorbitant, especially for high-resolution RS images. Moreover, a trained model typically performs outstandingly on modeling categories that are defined in training data but fails to detect any objects of new unseen categories. Given that RS images from different areas contain different objects—for instance, boats are hard to find in urban scene but are common in harbors—finding ways to adapt model to new unseen categories with affordable data costs is meaningful.

Few-shot learning has emerged as the predominant solution to such issues, which can adapt the model to new categories with a few support data by meta-learning strategy. During meta-training, a significant number of similar few-shot tasks are formatted to train the model, enabling it to adapt to a new task during meta-testing. Few-shot learning has been thoroughly studied in natural images [10], [11], [12]. The metric-based learning framework is prevalent in few-shot segmentation due to its efficiency and simplicity. Specifically, a transferable metric space is established to describe feature affinity, where similar features are compact, and discrepant features are distanced from each other, and then the target can be segmented according to the feature distance in the metric space. For example, PANet [13] and CANet [14] solve target segmentation by discriminating the distance between query features and class-specific prototype representations of support set in the metric space. HSNet [11] and BAM [12] built a metric space based on the cosine distance among each feature embedding, acquiring more fine-grained correlation map and better results. API [15] proposes probabilistic latent variable to represent prototype, which enhances the model's generalization ability. Recently, transformers have been widely implemented in computer vision to boost performance, so some studies also extend transformer to few-shot segmentation. For instance, Shi et al. [16] implement cross-attention modules to calculate similarities between support and query feature embeddings, while Hu et al. [17] proposed a transformer-based feature encoder to increase the intraclass feature compactness. Although few-shot learning has achieved remarkable success in natural images, there are few studies on few-shot learning for RS images.

The significant intraclass variance poses the primary obstacle to processing RS images. Due to the different capture conditions, geographic objects exhibit various appearances, scales, and perspectives as shown in Fig. 1. The intraclass variance hinders few-shot learning since the feature embeddings of the same semantic class may be separated from each other in the metric space, making it difficult to identify target objects

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Jing Wang, Qiang Zhou, Zhibin Wang, and Fan Wang are with the DAMO Academy, Alibaba Group, Hangzhou 310030, China (e-mail: Yunfei.wj@alibaba-inc.com; jianchong.zq@alibaba-inc.com; zhibin.waz@alibaba-inc.com; fan.w@alibaba-inc.com).

Yuang Liu is with State Grid Digital Technology Holding Company Ltd., Beijing 100077, China (e-mail: frankliu624@gmail.com).

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Fig. 1. Visualization of (a) ship and (b) swimming pool in different scales or appearances, and some objects in first column are highlighted by yellow boxes for convenient observation.

through prototype alignment or per-feature alignment. On the other hand, the presence of multiple types geometric objects with distinct distributions may split the parameter space and cause the model to extract class-specific features instead of a general feature descriptor for different tasks.

Recently, some works have attempted to address these issues, and the DMML-Net [18] proposed a deep feature pyramid comparison network for multiscale metric learning and introduced an affinity-based fusion mechanism to calibrating the deviation of prototypes. Similarly, SDM [19] adopted a gaussian mixture model to extract multiple prototypes, providing more accurate parsing, [20] adopt a background learner and contrastive learning to discriminate foreground from background, and [21] uses a nonparametric metric learning-based loss function to maximize the intraclass similarity of prototypes while minimizing the interclass similarity. However, all these mentioned methods directly implement global average pooling (GAP) to extract class-specific prototypes, which seriously compress the spatial information and performs poorly in eliminating the intraclass variance.

In this article, we present a fresh perspective on few-shot learning for RS images semantic segmentation. We partition few-shot feature matching into two sections—high-level semantic matching and low-level appearance matching. For high-level semantic matching, we assume that prototypes of each category follow the same distribution. Therefore, we initially adopt variational autoencoder (VAE) to create a learnable prototype module (LPM). It functions effectively in modeling the distribution of support features while eliminating the low-level variance like objects' appearance and scales. Specifically, we select ROIs from support feature embeddings according to the support ground-truth mask and send these feature embeddings to LPM to model prototype distributions. Comparing to GAP who compresses the feature embeddings into a vector to serve as prototype, the distribution learned by VAE works more efficiently in extracting high-level semantic representation. For low-level appearance matching, we propose a global attention-based correlation mapping module to achieve fine-grained feature alignment. In this instance, the global similarity of feature embeddings serve as the attention map, and their distance with GAP-based prototypes in the metric space will induce low-level features matching. Additionally, we implement a base-learner to alleviate the base class overfitting issue. Given the limited number of base categories in the training set and their large interclass distribution discrepancy, the meta-training model is prone to overfitting on base categories. Refer to BAM [12], we also trained an additional base learner to recognize objects of base classes and comprise it with the main branch (i.e., meta learner) to reduce the false predictions.

We evaluate our model on the public RS dataset iSAID-5^i, specially designed for RS few-shot semantic segmentation task [19]. The results indicate that our model set a new strong baseline for few-shot semantic segmentation in RS images.

Our main contributions are summarized as below.

1) We propose a VAE-based module to model the distribution of prototypes instead of nonparameterized GAP, and it works efficiently in extracting high-level class-specific semantic prototypes and eliminating the intraclass variance.
2) A global attention-based correlation mapping method is proposed to achieve low-level feature matching to assist accurate segmentation and prevent the omission of small-scale objects.
3) To mitigate overfitting, a base-learner is built to recognize objects belong to base categories and provide prior for the meta learner.

II. Related Works

In this section, we will introduce those related works about few-shot semantic segmentation on natural images and the expansion on RS images.

A. Few-Shot Semantic Segmentation

Few-shot has been widely used in image semantic segmentation as it is an effective method to adapt learned model to unseen categories with a few support samples. For example, Shaban et al. [22] proposed the pioneering few-shot segmentation model, where a condition branch is built to produce a set of parameters from labeled support samples and these parameters work as classifier to classify query features extracted from another branch. Similarly, Rakelly et al. [23] also use a condition branch to make a globally pooled prediction that informs the segmentation of the query from support set, and the predictions are concatenated with the query feats for final segmentation. Dong and Xing [10] proposed the concept of prototype; they extracted prototype for a specific category by GAP layer and the query image is segmented according to the l2-distance between query features and corresponding class prototype. Due to the effectiveness of prototype, a lot of methods implement prototype to assist the few-shot learning. Zhang et al. [24] segment query images by calculating the cosine distance between query features and class prototype, Liu et al. [25] decompose the holistic class representation into a set of part-aware prototype to capture diverse

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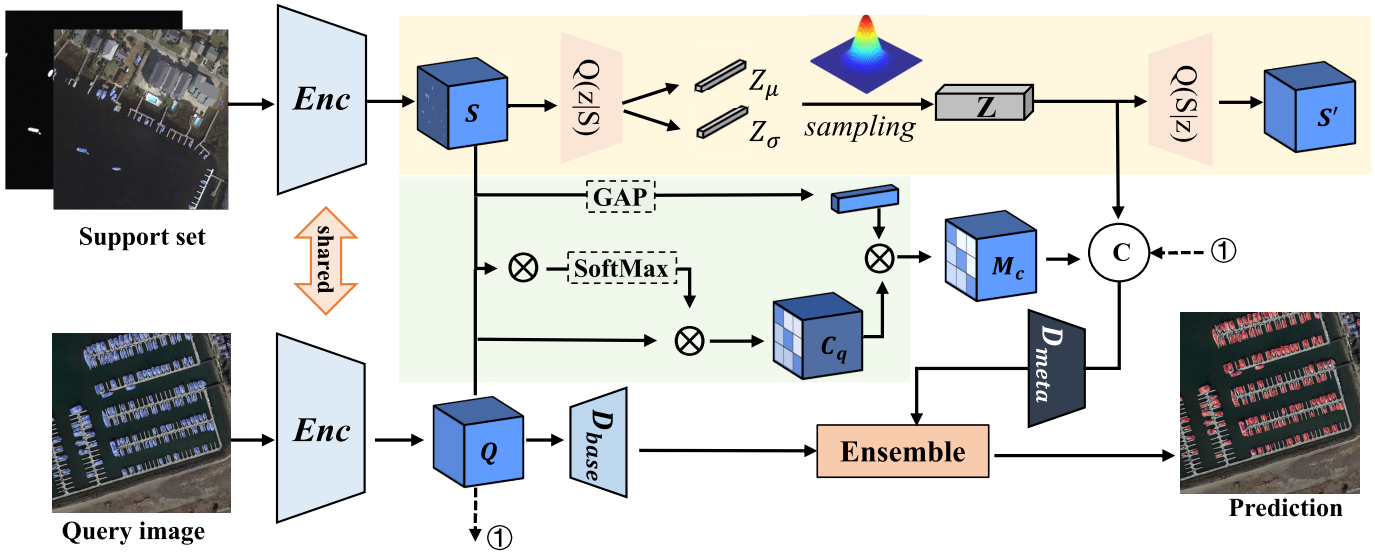


Fig. 2. Overall architecture of the proposed method, which consisted of feature encoder (Enc), variational prototype extraction (light yellow part), global attention-based correlation mapping module (light green part), and two decoders (D_{meta} and D_{base}).

and fine-grained object features, Yang et al. [26] proposed an expectation–maximization algorithm to extract multiple prototypes from support set, and Wang et al. [13] improve the segmentation accuracy by prototype alignment through a cycle-training manner. Recently, Liu et al. [25] proposed an adaptive prototype learning method where multirepresentative prototypes can be extracted by aggregating similar support feature vectors. Lang et al. [27] proposed a divide-and-conquer proxies where the support image was divided into multiple regions and assigned a specific prototype to each region. To achieve more fine-grained correlation from support features and query features, Min et al. [11] extract a multilevel correlation tensors to represent support-query correspondence. Moreover, to solve the obstinate overfitting problem, Lang et al. [12] proposed a base-learner to identify objects of base classes to address the bias problem.

B. Few-Shot Learning in RS Images

Few-shot learning has been widely studied in RS image classification or geographic objects detection due to its low cost. For example, Li et al. [28] introduce attention technique to discover the discriminative regions to boost scene classification accuracy, Cheng et al. [29] designed a Siamese-prototype network with prototype self-calibration and intercalibration to extract valid prototypes inducing the scene classification of query samples, and Zhang et al. [30] process images into scene graph and a scene graph matching module is further developed to evaluate the similarity score between each image and each scene class. Moreover, Zeng and Geng [31] and Li et al. [32] plug the popular self-training method in few-shot RS images classification. To better interpret the images, some studies figure out objects detection by few-shot learning; Li et al. [33] propose a feature reweighting module to learn class-specific reweighting vectors from support features and calibrate the query features according to these vectors to achieve objects detection, similarly, Cheng et al. [34] propose a prototype learning module to extract class-aware prototypes and guide the region proposals generation of query images.

C. Few-Shot Semantic Segmentation in RS Images

Semantic segmentation is important for analyzing the RS images, and many effective works have been proposed [35], [36], [37]. To solve the recognition of novel objects, few-shot semantic segmentation has been introduced to RS images recently. Wang et al. [18] propose a few-shot segmentation model by combining a meta-metric learner and a base-metric learner to segment geographic objects in RS images, where both metric learners are built by deep feature pyramid comparison network. Besides, Yao et al. [19] proposed a scale-aware few-shot semantic segmentation network to perform detailed matching with multiple prototypes. Comparing to the remarkable achievements in natural images, there is great potential in improving the performance of few-shot semantic segmentation in RS images.

III. Methods

A. Problem Setting

In this article, we aim to segment previously unseen geographic objects in RS images during evaluation with a few labeled prior examples. A few-shot learning framework is plugged in to build the proposed model. The dataset is split in training set D_{train} and test set D_{test}, where the categories in train set (C_{train}) and those in test set (C_{test}) are nonoverlapping in few-shot training. The model is trained in an episode manner, where a support set S contained K labeled images of class c and a query set Q of class c are needed in each training cycle, and the model learns to predict mask I_q for $\mathbf{M}_q$. Here, we denote the support set and query set as $\{(I_{\text{sk}}, \mathbf{M}_{\text{sk}})\}_{k=1}^{K}$ and $\{(I_q, \mathbf{M}_q)\}$, respectively, where $I \in \mathbb{R}^{H\times W\times 3}$ represents image and $\mathbf{M} \in \{0, 1\}^{H\times W}$ is corresponding mask. After training, the model is able to detect new categories objects in D_{test} when given a small corresponding support set.

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B. Overall Architecture

As shown in Fig. 2, the proposed model contains three main components: a backbone for feature extraction, correlation mapping for few-shot learning, and decoder for final prediction.

During training and testing, we first feed the support set and query set to the pretrained convolution neural networks (CNNs) like VGG16 [38] and Resnet50 [39] to extract class-agnostic features **S** and **Q**. Second, we use a variational autoencoder (VAE) to learn the distributions of prototypes $\mathbb{N}$ from the masked support image features. At the same time, a global attention-based correlation mapping module are leveraged on the masked support features and query features to generate fine-grained correlation map. The correlation map, along with prototypes sampled from corresponding distributions $\mathbb{N}$, are concatenated with mid-level query feature and fed into the meta-decoder to generate meta-predictions. Moreover, the few-shot model is easy to bias toward base classes of training data due to the unbalanced data distribution and large domain shift, and here, we implement a base-learner [12], [40], [41] to mitigate this bias problem. Specifically, the base-decoder explicitly detects base classes that should not be detected in the query images. Finally, the meta-predictions and the base-predictions are combined using an ensemble layer to obtain the final predictions.

C. Variational Prototype Extraction

GAP is the most common method to acquire class-specific prototype in few-shot semantic segmentation [10], [25], which compresses the region of interest features into a single or multi vectors and count these vectors as prototypes, formulated as follows:

$$P_s^c = \frac{\sum_{x,y} \mathbf{S}^{x,y} \mathbb{I}\left[\mathbf{M}_s^{x,y} = c\right]}{\sum_{x,y} \mathbb{I}\left[\mathbf{M}_s^{x,y} = c\right]} \tag{1}$$

where c represents the specified category, $\mathbf{S}^{x,y}$ represents the support feature located at (x, y), and $\mathbb{I}$ is an indicator function, outputting value 1 if argument is true or 0 otherwise.

Even the GAP-based prototype extraction has achieved great success, the prototype vector is inadequate to represent the distribution of a whole category in RS images. The intraclass variance is unable to be ignored as well as the tiny objects in RS images. In this article, we use a VAE-based module to predict a distribution to represent a specific category instead of a vector. The difference between the prototype vector and prototype distribution are displayed in Fig. 3, it is obvious that a distribution will be more representative and softer than a single vector in high-dimension space.

Formally, we extracted the support ROI features by support masks: $\mathbf{S} = \mathbf{S}_{\text{all}} \odot \mathbf{M}_s$, and use a VAE to predict the class distribution $\mathbb{N}$ from the ROI support features S, then sampling prototypes from $\mathbb{N}$ to guide further object segmentation. A VAE generally consisted of an encoder F_{enc} and a decoder F_{dec}; F_{enc} aims to predict a variational approximate posterior $Q(z|\mathbf{S})$ for an input sample, then F_{dec} learns to reconstruct the input from latent variable Z that sampled from the $Q(z|\mathbf{S})$.

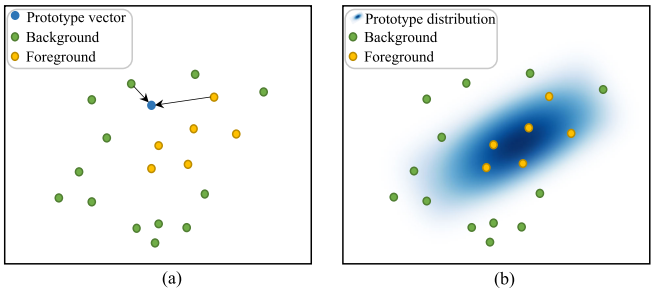


Fig. 3. GAP-based (a) prototype vector and (b) prototype distribution in hidden space.

Here, we use the encoder F_{enc} of VAE to predict the distribution of specific class $Q(z|\mathbf{S})$ from the masked support features, whose true distribution is assumed as $P(z|\mathbf{S})$, and we optimize the model by minimizing their Kullback–Leibler (KL) divergence

$$D_{\text{KL}}(Q(z|\mathbf{S}) \| P(z|\mathbf{S})) = \mathbb{E}_{z\sim Q}\left[\log Q(z|\mathbf{S}) - \log P(z|\mathbf{S})\right] \tag{2}$$

and according to the Bayes rules, (1) can be written as

$$D_{\text{KL}}(Q(z|\mathbf{S}) \| P(z|\mathbf{S})) = \mathbb{E}_{z\sim Q}\left[\log Q(z|\mathbf{S}) - \log P(\mathbf{S}|z) - \log P(z)\right] + \log P(\mathbf{S}) \tag{3}$$

which equals to

$$\log P(\mathbf{S}) - D_{\text{KL}}[Q(z|\mathbf{S}) \| P(z|\mathbf{S})] = \mathbb{E}_{z\sim Q}\left[\log P(\mathbf{S}|z)\right] - D_{\text{KL}}[Q(z|\mathbf{S}) \| P(z)]. \tag{4}$$

The right part of (3) is the cost function of VAE, i.e., evidence lower bound (ELBO) [42]. We assume that prior distribution of z follows a standard Gaussian, $P(z|\mathbf{S}) \sim \mathbb{N}(0, I)$, the approximate posterior distribution $Q(z|\mathbf{S})$ is a multivariant Gaussian with diagonal covariance $Q(z|\mathbf{S}) \sim \mathbb{N}(\mu, \sigma)$. Then, the objects of F_{enc} is to predict the value of μ and σ according to the input support features $\mu, \sigma = F_{\text{enc}}(\mathbf{S})$. Following the implementation of VAE, we use the reparameterization trick to sample variational prototype z,

$$z = \mu + \sigma \cdot \epsilon \tag{5}$$

where $\epsilon \sim \mathbb{N}(0, 1)$. Then, the first term of (3) can be cast into calculating the loss between the input **S** and the reconstructed **S'**, and here we implement L2 loss

$$L_{\text{rec}} = \|\mathbf{S} - \mathbf{S'}\|_2 \tag{6}$$

where $S' = F_{\text{dec}}(z)$. And the second term of (3) is optimized by minimizing the KL divergence of $Q(z|\mathbf{S})$ and $P(z|\mathbf{S})$

$$L_{\text{KL}} = D_{\text{KL}}[Q(z|\mathbf{S}) \| P(z|\mathbf{S})]. \tag{7}$$

The whole loss for optimize this module is formulates as:

$$L_{\text{vae}} = L_{\text{KL}} + L_{\text{rec}}. \tag{8}$$

With the optimization of L_{vae}, support features **S** will be transferred into a distribution $\mathbb{N}$, as only features belonging to specific class will be reserved with the guidance of support masks, and the distribution can be recognized as the prototype distribution of the class. Then, we sample variational feature z from $\mathbb{N}$ to serve as the class prototype.

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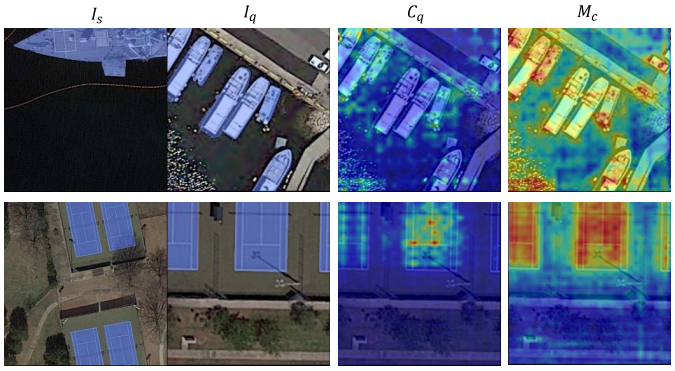


Fig. 4. Visualizations of global attention map $\mathbf{C}_q$ and our GACM $\mathbf{M}_c$.

D. Global Attention-Based Correlation Map

The prototype solves the feature alignment issue in semantic level, and we further proposed a global attention-based mapping module to help the low-level features mapping, as shown in Fig. 2 (the green part). Cosine similarity map is implemented widely to assist low-level feature mapping, Formally, the cosine similarity between the masked support features $\mathbf{S}$ and the whole query features $\mathbf{Q}$ will be calculated first

$$\mathbf{C}_{\text{qs}}(x^q, x^s) = \frac{\mathbf{Q}(x^q)^T \cdot \mathbf{S}(x^s)}{\|\mathbf{Q}(x^q)\| \|\mathbf{S}(x^s)\|} \tag{9}$$

where x^s and x^q represent 2-D spatial positions of support feature maps $\mathbf{S}$ and query feature maps $\mathbf{Q}$, respectively. Then, for each $x^q \in \mathbf{Q}$, the similarity score along all support pixels will summed as the global attention value

$$\mathbf{C}_q = \sum_{s=1}^{\text{hw}} \texttt{softmax}(\mathbf{C}_{\text{qs}}) \tag{10}$$

where hw is the spatial size of support feature map. But, we found this global attention map is not suitable for RS images due to large intracategory divergence. As shown in Fig. 4, when appearance of object in support image is different from that in query image (the first row), the $\mathbf{C}_q$ fails to focus on exact objects. Similarly, objects will be neglected when it is partly sheltered (the second row). Under such conditions, $\mathbf{C}_q$ is unable to provide fine-grained instruction. So, we further improve $\mathbf{C}_q$ to dig deeper correlation between support image and query image.

First, we multiply the attention map with the query features to initially remove the prominent unrelated part, e.g., background, formulated as:

$$\hat{\mathbf{Q}} = \mathbf{C}_q \odot \mathbf{Q} \tag{11}$$

where $\odot$ represents Hadamard product. Then to further dig correlation between support set and query image and acquire the fine-grained correlation map $\mathbf{M}_c$, we calculate the mapping score between the GAP-based support prototype $\mathbf{P}_s$ and $\hat{\mathbf{Q}}$

$$\mathbf{M}_c(x_q) = \frac{\mathbf{P}_s{}^T \cdot \hat{\mathbf{Q}}(x^q)}{\|\mathbf{P}_s\| \|\hat{\mathbf{Q}}(x^q)\|} \tag{12}$$

where P_s is calculated by (1).

The M_c is shown in Fig. 4, and it works effectively in increasing focus for exceptive objects and provide richer information. In this work, we assign M_c to work as guidance for low-level features alignment.

E. Base Learner Assistance

As we train the model in meta-learning manner, the model is easy to bias toward base classes due to the unbalanced data, and it impedes the recognition of novel categories. In our proposed model, we implement a base learner to mitigate this issue, who explicitly detects base classes that should not be detected during evaluation. The base learner shares backbone network with mata-learner but possesses a personal decoder D_{base} to recognize objects that belong to base categories, as shown in Fig. 2. We train the D_{base} follow a supervised manner instead of meta-learning. Specifically, the query image is fed to backbone first to extract features $\mathbf{Q} \in \mathbb{R}^{c \times h \times w}$, then the D_{base} consists of several convolution blocks and downsample layers will process $\mathbf{Q}$ and yields the final prediction, which can be defined as

$$\hat{y}_b = \texttt{softmax}(D_{\text{base}}(\mathbf{Q})) \in \mathbb{R}^{(1+N_b) \times H \times W} \tag{13}$$

softmax$(\cdot)$ operation is conducted along channel dimension to generate the base classes predictions $\hat{y}_b$, N_b represents the number of base categories. Similar to normal supervised training, the base-learner is optimized through the cross-entropy (CE) loss between the prediction $\hat{y}_b$ and the ground-truth of base classes $\mathbf{M}_{\text{qb}}$.

We train the whole model in a two-stage training strategy, the base learner will be trained first, and then freezing it when training the meta-learner. For the meta training branch, we combine the query features $\mathbf{Q}$ with the variational prototype z and fine-grained correlation map $\mathbf{M}_c$ to get the meta predictions $\hat{y}_m$, defined as

$$\hat{y}_m = \texttt{softmax}(D_{\text{meta}}(\mathbf{Q}, z, \mathbf{M}_c)), \quad \hat{y}_m \in \mathbb{R}^{(1+1) \times H \times W} \tag{14}$$

where D_{meta} denotes the meta decoder, which contains a concatenate operation and three convolution blocks. Each convolution block includes a 3×3 convolution layer and a ReLU activation layer, which will detect all objects of the specified novel category. The binary cross entropy (BCE) is implemented to update the meta learner

$$L_{\text{meta}} = L_{\text{BCE}}(\hat{y}_m, \mathbf{M}_q) \tag{15}$$

where $\mathbf{M}_q$ denotes the ground-truth masks of novel class objects.

The prediction of D_{meta} is shown in Fig. 5, where $\hat{y_m}$ represents the prediction of it; 0 and 1 represent background and foreground respectively. It is natural that the small vehicles (highlighted by yellow arrow) are segmented in the ship segmentation task as D_{meta} is biased on base classes due to limited support data. To solve this issue, we ensemble the meta and base output to acquire the final predictions. Specifically, we combine the foreground predicted by D_{base} and background

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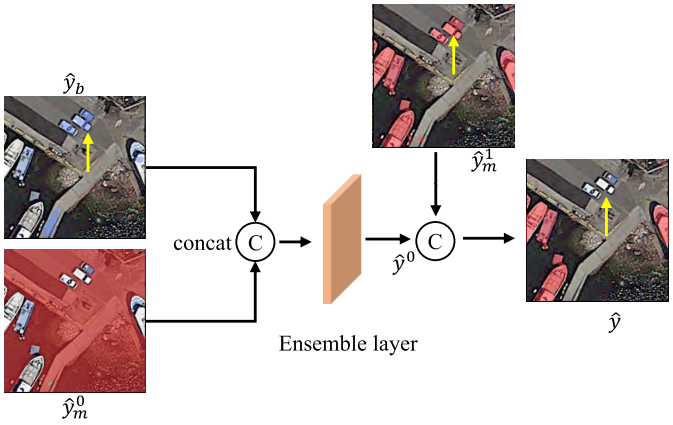


Fig. 5. Base learner assistance process, $\hat{y}_m$ and $\hat{y}_b$ are the output of meta decoder and base decoder, respectively. The meta decoder is easy to misidentify car (base class) as ship (novel class), and we remove cars from the final prediction according to the prediction of base learner ($\hat{y}_b$). The ensemble layer contains a convolution layer and a ReLU activation layer.

predicted by D_{meta} to form the predicted background, defined as

$$\hat{y}^0 = F_{\text{en}}\left(\left(\sum_{i=1}^{N_b} \hat{y}_b^1\right), \hat{y}_m^0\right) \tag{16}$$

where F_{en} is the ensemble layer consisted of a 1×1 convolution and a ReLU layer, which will encode the new background features. Then, the new background representation is concatenated with the $\hat{y}_m^1$ to get the final predictions

$$\hat{y} = \hat{y}^0 \oplus \hat{y}_m^1 \tag{17}$$

where $\oplus$ means channel-wise concatenation. The prediction loss is calculated by

$$L_{\text{pred}} = L_{\text{BCE}}\left(\hat{y}, \mathbf{M}_q\right). \tag{18}$$

The overall objectives of the framework is formulated as:

$$L = L_{\text{pred}} + L_{\text{meta}} + L_{\text{vae}}. \tag{19}$$

IV. Experiments

In this section, we shall begin by acquainting ourselves with the dataset that we employ to evaluate our method. Subsequently, we shall delineate the details of our experiments, and culminate our discussion by presenting the experimental results and comparing them with state-of-the-art methods.

A. Datasets

We utilized the public iSAID-5^i dataset as a benchmark to assess the efficacy of our method. The iSAID-5^i is a recently introduced dataset intended for few-shot semantic segmentation of RS images [19], which was derived from the extensive instance segmentation dataset iSAID [43], which comprised of 2806 high-resolution aerial images with a total of 655 451 object instances across 15 categories. The iSAID was adapted for few-shot semantic segmentation by Yao et al. [19], who restructured it based on the settings of Pascal-5^i, a frequently used few-shot segmentation dataset [22]. More specifically, the iSAID-5^i comprises 18 076 training images, 6363 validation images, and all images are cropped at 256×256. Furthermore, following [19], we divided the dataset into three folds, each consisting of five object categories. We employed cross-validation to evaluate the model's performance, with each fold serving as a test set in turn whilst the remaining two folds served as the training sets. Table I displays the object categories included in each fold.

TABLE I
Categories Contained in Each Fold

Fold	Test classes
0	ship, storage tank, baseball diamond, tennis court, basketball court
1	ground track field, bridge, large vehicle, small vehicle, helicopter
2	swimming pool, roundabout, soccer ball field, plane, harbor

B. Implementation Details

The proposed model is trained in two stages. In the first stage, we train the base learner in a standard supervised learning paradigm on the training data of each fold. The PSPNet is adopted to build the base learner, and we trained it for 100 epochs on iSAID-5^i with learning rate of 2e-4, batch size of 4, and SGD optimizer is implemented to update parameters. For the second stage, we train the other modules, dubbed as meta learner, using episodic learning manner while keeping parameters of the base learner fixed during this stage. Note that both the base and meta learner share the same feature encoder, and we selected VGG16 and ResNet50 that pretrained on ImageNet [45] as our feature encoder, which is also fixed during training to prevent overfitting. We employ SGD optimizer to update the networks' parameters, with an initial learning rate set to 1e-3 and a batch size of 16 for all datasets. The meta learner was trained for 200 epochs. An ASPP module and three convolution blocks make up the meta decoder. We employ SGD optimizer to update the networks' parameters, with an initial learning rate set to 1e-3 and a batch size of 16 for all datasets. The meta learner was trained for 200 epochs.

Similar to previous works, we evaluate model's performance by mean intersection-over-union (mIoU). And we follow [12] to expand the proposed method to k-shot setting, where a weight layer is added to assess the weight of each support image and then the prototypes generated from support images are added according to the weight matrix to form the final prototype.

C. Experimental Results and Comparison

We have compared our method with previous few-shot segmentation methods designed for both natural images and RS images, and our results on iSAID-5^i are displayed in Table II. It is evident that our method outperforms all previous methods in both one-shot and five-shot setting by a significant margin. To be more specific, under one-shot setting, we achieve a 14.79% and 17.35% mIoU improvement across all folds for VGG16 and Resnet50 backbone, respectively, compared to the SOTA FSS method on RS images, i.e., SDM. It is worth noting that our methods achieves more than 20% mIoU

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TABLE II

COMPARISON BETWEEN OUR METHOD AND PREVIOUS METHODS ON iSAID-5^i. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. THE UNDERLINE METHODS NEED FINE-TUNING MODEL ON THE SUPPORT SET

Backbone	Method	1-shot (mIoU)				5-shot (mIoU)			
		Fold-0	Fold-1	Fold-2	Mean	Fold-0	Fold-1	Fold-2	Mean
VGG16 [38]	PANet [13]	17.43	11.43	15.95	14.94	17.70	14.58	20.70	17.66
	CANet [14]	19.73	17.98	30.93	22.88	23.45	20.53	30.12	24.70
	PMMs [26]	20.87	16.07	24.65	20.53	23.31	16.61	27.43	22.45
	PFENet [44]	16.68	15.30	27.87	19.95	18.46	18.39	28.81	21.89
	SDM [19]	29.24	20.80	34.73	28.26	36.33	27.98	42.39	35.57
	Ours	**49.71**	**35.08**	**46.00**	**43.60**	**51.77**	**35.62**	**47.77**	**45.05**
ResNet50 [39]	PANet [13]	12.36	9.11	12.05	11.17	13.82	12.40	19.12	15.11
	CANet [14]	18.80	15.62	25.79	20.07	23.86	18.54	32.00	24.80
	PMMs [26]	19.02	18.51	28.42	21.98	20.89	20.87	31.23	24.33
	PFENet [44]	18.75	17.24	22.09	19.36	19.57	18.43	26.14	21.38
	API [15]	25.42	24.76	27.30	25.83	24.28	24.76	26.5	25.18
	DCAMA [16]	42.99	29.62	43.12	38.58	50.89	35.00	**51.79**	45.89
	SDM [19]	34.29	22.25	35.62	30.72	39.88	30.59	45.70	38.72
	Ours	**55.32**	**39.25**	**49.64**	**48.07**	**56.82**	**41.92**	51.30	**50.01**

TABLE III

MIOU RESULTS FOR ALL CATEGORIES IN iSAID-5^i UNDER ONE-SHOT SETTING, C1: SHIP, C2: STORAGE TANK, C3: BASEBALL DIAMOND, C4: TENNIS COURT, C5: BASKETBALL COURT, C6: GROUND TRACK FIELD, C7: BRIDGE, C8: LARGE VEHICLE, C9: SMALL VEHICLE, C10: HELICOPTER, C11: SWIMMING POOL, C12: ROUNDABOUT, C13: SOCCER BALL FIELD, C14: PLANE, AND C15: HARBOR. RESULTS OF PREVIOUS METHODS ARE QUOTED FROM SDM DIRECTLY. THE BEST RESULTS ARE IN BOLD

Method	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	Mean
VGG16 [38]																
PANet [13]	13.85	17.78	19.78	17.80	17.94	22.41	11.75	11.75	6.36	4.88	12.96	19.11	23.49	11.87	12.33	14.94
CANet [14]	7.51	12.87	31.22	30.45	16.61	18.31	31.94	23.15	3.77	12.55	33.99	**51.41**	36.69	17.61	14.91	22.87
PMMs [26]	12.84	23.06	27.81	26.65	13.97	18.48	13.76	25.58	11.68	10.88	28.52	34.62	34.15	16.32	9.64	20.53
PFENet [44]	3.94	4.70	30.95	29.21	14.59	15.01	14.63	27.68	10.13	9.06	29.40	47.89	25.23	25.20	11.63	19.95
SDM [19]	26.55	30.57	33.01	35.06	21.03	20.54	28.89	32.45	9.62	12.59	30.79	43.52	49.47	27.51	**22.37**	28.26
Ours	**42.41**	**60.56**	**47.41**	**58.46**	**39.69**	**35.62**	**48.02**	**44.59**	**24.25**	**22.94**	**50.62**	50.20	**61.46**	**56.74**	10.98	**43.60**
ResNet50 [39]																
PANet [13]	7.27	9.99	14.49	13.2	16.85	17.40	10.23	7.37	6.21	4.33	8.35	12.38	21.28	8.52	9.75	11.17
CANet [14]	7.25	27.25	43.32	6.17	9.99	6.16	17.81	28.58	8.84	16.73	17.73	56.97	13.82	27.41	13.06	20.07
PMMs [26]	12.93	17.92	41.97	7.43	14.87	13.24	32.67	26.08	4.17	16.37	12.26	**61.49**	38.37	17.92	12.02	21.98
PFENet [44]	2.98	6.30	33.29	34.49	16.71	8.68	8.67	34.62	18.19	16.08	28.58	24.47	7.37	39.76	10.31	19.37
SDM [19]	37.66	34.37	34.45	39.81	25.14	16.77	34.53	30.50	12.42	17.02	20.69	56.83	42.80	40.52	17.26	30.72
Ours	**53.61**	**66.41**	**54.79**	**60.55**	**41.24**	**32.22**	**43.36**	**53.43**	**29.61**	**37.63**	**50.62**	58.96	**58.54**	**57.61**	**22.45**	**48.07**

improvement on Fold-0. For the five-shot setting, our method consistently yields significant performance improvement compared to SDM, with a 9.48% and 11.29% mIoU increasing for VGG16 and ResNet50 backbones, respectively. However, we observe that the five-shot performance of our method only improved slightly compared to the one-shot setting. We assume that is because, in multishot context, the prototype learning module needs to generate multiple prototypes first and then combine them together to formulate a new common prototype, which may be too abstract to achieve semantic alignment with query features.

The mIoU results for all categories are displayed in Table III. It is obvious that our method demonstrate excellent performance in segmenting different kinds of geographic objects; for example, we improved more than 30% mIoU on small-scale object like storage tank (C2) and swimming pool (C11), and more than 20% mIoU on large-scale objects like baseball diamond (C3), tennis court (C4), and plane (C14) compare to SDM. All these results demonstrate the effectiveness of the proposed method.

TABLE IV

ABLATION STUDY RESULTS ON EACH COMPONENT OF THE PROPOSED MODEL

Module			mIoU
GAP prototype	VAE prototype	GACM	
✓			48.37
✓		✓	50.89
	✓		48.46
		✓	52.56
	✓	✓	**55.32**

D. Ablation Study

We conduct ablation studies to demonstrate how the prototype learning and global attention-based correlation map (GACM) modules affect our model. Note that all experiments in this section are performed on Fold-0 data with ResNet50 backbone, unless specified otherwise. The results are presented in Table IV. During the ablation study, we concatenate the

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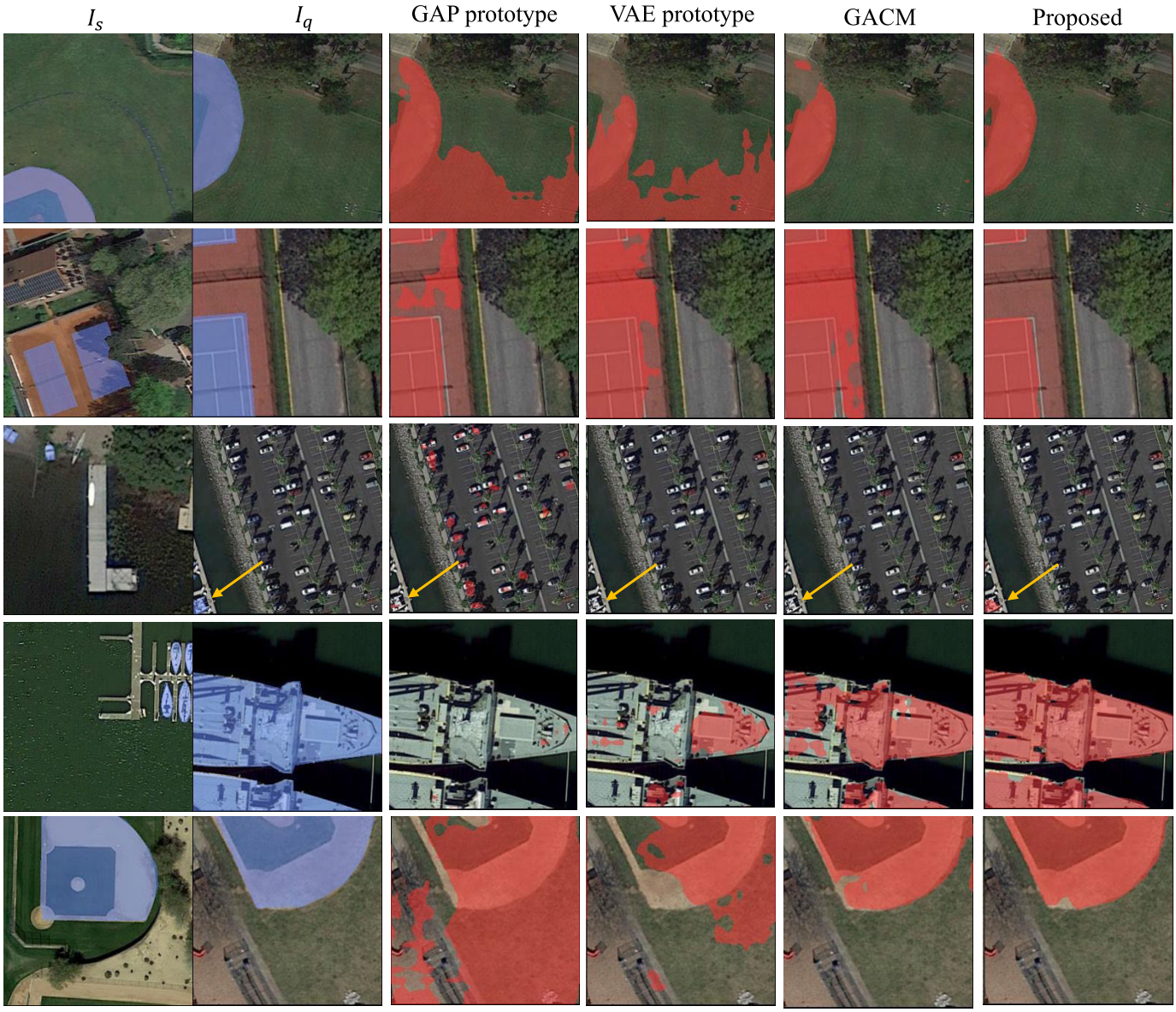


Fig. 6. Qualitative results of different modules. The first two columns are input support and query images with objects masked by light blue, the segmentation results of models with only GAP prototype, VAE prototype, and GACM as prompts are displayed in the following columns, respectively, and results of the proposed model are displayed in the final column. Segmentation masks are in light red color. Yellow arrows are employed to highlight targets.

prompts with query features and feed them to the decoder to obtain segmentation results. First, we set a baseline where only GAP-based prototype are provided as prompt, and it achieves a mIoU of 48.37%. Then, we add the GACM as a prompt, and the mIoU performance improved by 2.52%. When replacing the GAP-based prototype with the proposed variational prototype (the third row in Table IV), the model's performance improved slightly (i.e., 0.09%). When setting the prompt as the GACM (the forth row of Table IV), model improved its performance by 4.1% compared to using the VAE prototype alone or 1.67% higher than combining the GACM with GAP-based prototype (the second row of Table IV). Finally, we combine the VAE with the GACM, and the model achieves the best result with mIoU of 55.32%. According to these results, we find that both the VAE prototype and GACM works effectively in providing prompt for target segmentation, and combining them will lead to better results. It is worth noticing that combining GAP-based prototype with the GACM brings performance drop compared to GACM alone; we infer that because the GAP prototype provides a global semantic prompt while the GACM provide a local prompt, where both of them are sensitive to appearance, objects scale, and shape-variety of geographic objects, which lead to inconsistency among them. The variational prototype further learns the distribution of the target categories, which remedies the correlation between objects with quite different appearance. Therefore, combining it with GACM achieves better results.

E. Qualitative Results

To better observe the excellence of our model, we visualize the segmentation results in Fig. 6. First, we plot the results of model that used only GAP prototype as prompt, it is clear that

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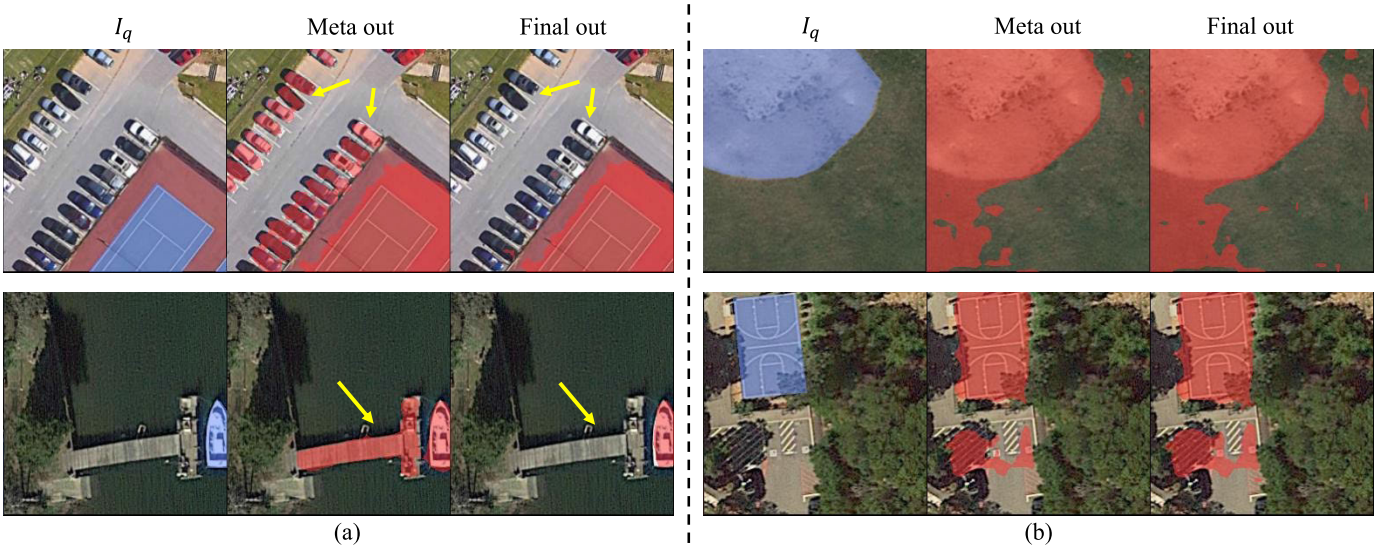


Fig. 7. Qualitative results of meta decoder output and the final output of our method. (a) Display the successful samples where base classes objects (indicated by yellow arrays) are removed. (b) Displays samples that objects out of base classes cannot be detected by base detector.

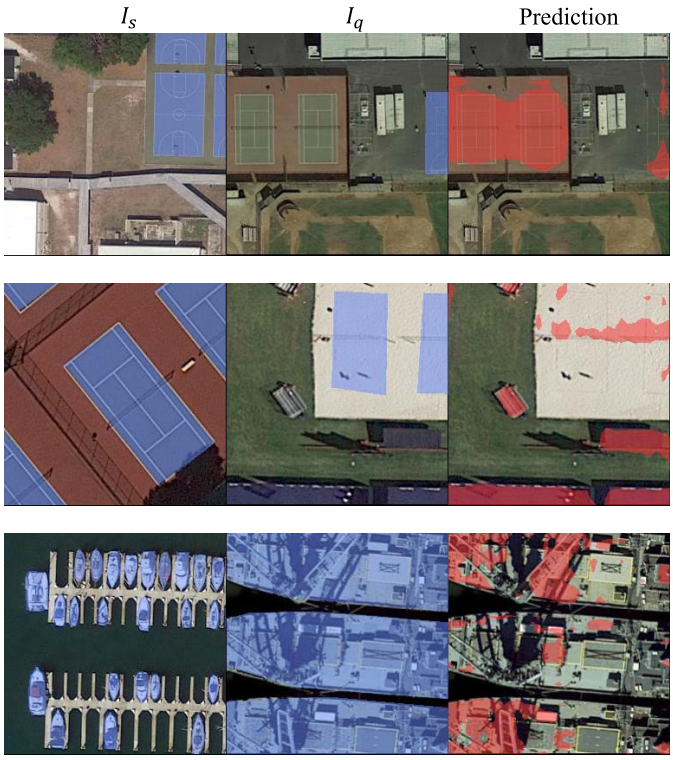


Fig. 8. Failure cases, the ground-truth masks are highlighted by light blue and the prediction masks are in light red. The first row displays that the model is hard to discriminating tennis courts and basketball field. The second row displays that the model cannot match objects with quite different appearance. The last row displays that the model cannot detect ships that are occluded heavily.

the model focuses on background that has similar texture to target objects (the first two rows), misidentifies small vehicles as ship because of similar appearance (the third row), and fails to detect huge ship in query image due to scale differences with small ship in support image (the final row). The results of third column demonstrate GAP prototype is sensitive to objects' appearance and with narrow intercategories gap. After replacing GAP prototype with VAE prototype, the model still misidentifies background as basketball field or tennis court, but cars are separated with ships, and huge ships can be partly recognized with the instruction of small ship. These results proved that the VAE prototype can reduce intracategories discrepancy and increase intercategories gap. As prototype is a kind of general prompt which will leave out a lot of details, we implement the pixel-to-pixel prompt, i.e., GACM to replace it. As shown in fifth column of Fig. 6, some misidentified background in former two models are corrected (the first row), and more parts of huge ship are detected (the final row), which demonstrate that pixel-to-pixel GACM can provide more delicate instruction. Finally, we display the segmentation results of our model, and it is clear that all objects are correctly segmented regardless of appearance variety or interclasses similarity. These results further prove the effectiveness of our method.

Moreover, to solve the overfitting issue in few-shot learning, we introduce a base decoder to remove misidentified base classes objects from the final segmentation masks. To understand this process more intuitively, we displayed the outputs of meta decoder (meta out) and final ensemble results (final out) in Fig. 7, where (a) displays the successful example where base classes objects like cars (first row) and harbors (second row) are removed from the meta out to form the final out, and (b) displays extra sample to demonstrate that base decoder can only help to remove objects of base classes so that misidentification of objects like lawn (first row) and playground (second row) cannot be revised. Even the proposed method gains satisfactory results, but there still are a few difficult cases it cannot handle, mainly lie in: 1) discriminating similar classes like basketball field and tennis court; 2) matching objects with quite different appearance; and 3) detecting heavily occluded objects, as shown in Fig. 8.

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V. Conclusion and Discussion

In this article, we propose an efficient few-shot semantic segmentation model to solve the RS images segmentation, where the model can adapt to a new geographic objects with a few labeled support images. To mitigate the appearance diversity of RS images, we design a LPM to capture high-level semantic representation and a global-attention-based correlation matching module to implement fine-grained feature matching. The results on the public RS datasets demonstrate the validity of our model, and we set a new baseline for few-shot semantic segmentation of RS images.

However, in addition to the amazing results, we found there are a few issues needing further research. First, we found that the proposed model is bad at segmenting thin and small objects like harbor and helicopter, which is a stubborn problem in RS image segmentation. Second, as we use a distribution to depict a category, some semantic-close classes objects like small vehicle and large vehicle are hard to distinguish. Third, for objects without depth difference from their background like tennis court and basketball court et al., the model is unable to segment out their edge accurately, introducing some extra information like spectral map might improve. Also, with the emergence of the general instance segmentor, SAM, it is worth exploring how to combine it with few-shot semantic segmentation to acquire better performance. We will keep studying on these issues in the future.

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Jing Wang received the Ph.D. degree from the University of Science and Technology of China, Hefei, China, in 2022.

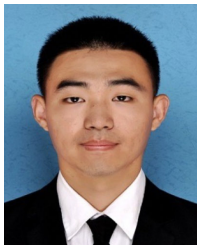
She is an Engineer with Alibaba Group, Hangzhou, China. Her research interests include computer vision, few-shot learning, and weakly supervised learning.



Yuang Liu received the B.E. degree in software engineering from Wuhan University of Science and Technology, Wuhan, China, in 2019. He is currently pursuing the Ph.D. degree with the School of Computer Science and Technology, East China Normal University, Shanghai, China.

His research interests include computer vision, model compression, and knowledge distillation. He has been invited as a reviewer for top international conferences and journals, including CVPR, ICCV, ECCV, ICLR, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE TRANSACTIONS ON INTELLIGENT VEHICLES, and *Neurocomputing*.

Qiang Zhou is an Engineer with Alibaba Group, Hangzhou, China. His research interests include deep learning, computer vision, and 3-D reconstruction.



Zhibin Wang is a Staff Engineer with Alibaba Group, Hangzhou, China. His research interests include deep learning, remote sensing, and computer vision.

Fan Wang is a Senior Staff Engineer with Alibaba Group, Hangzhou, China.

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